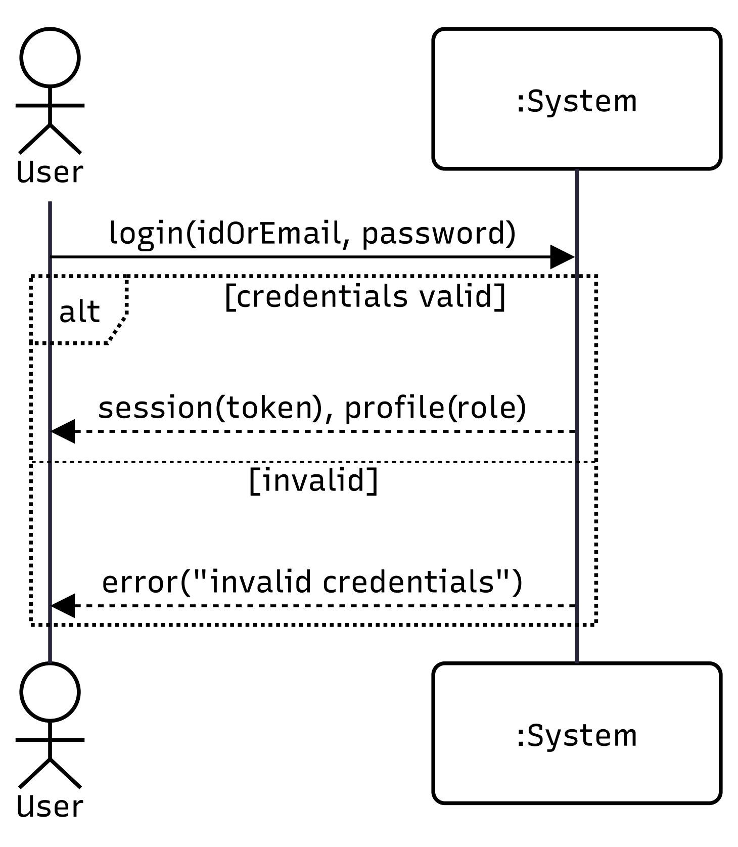
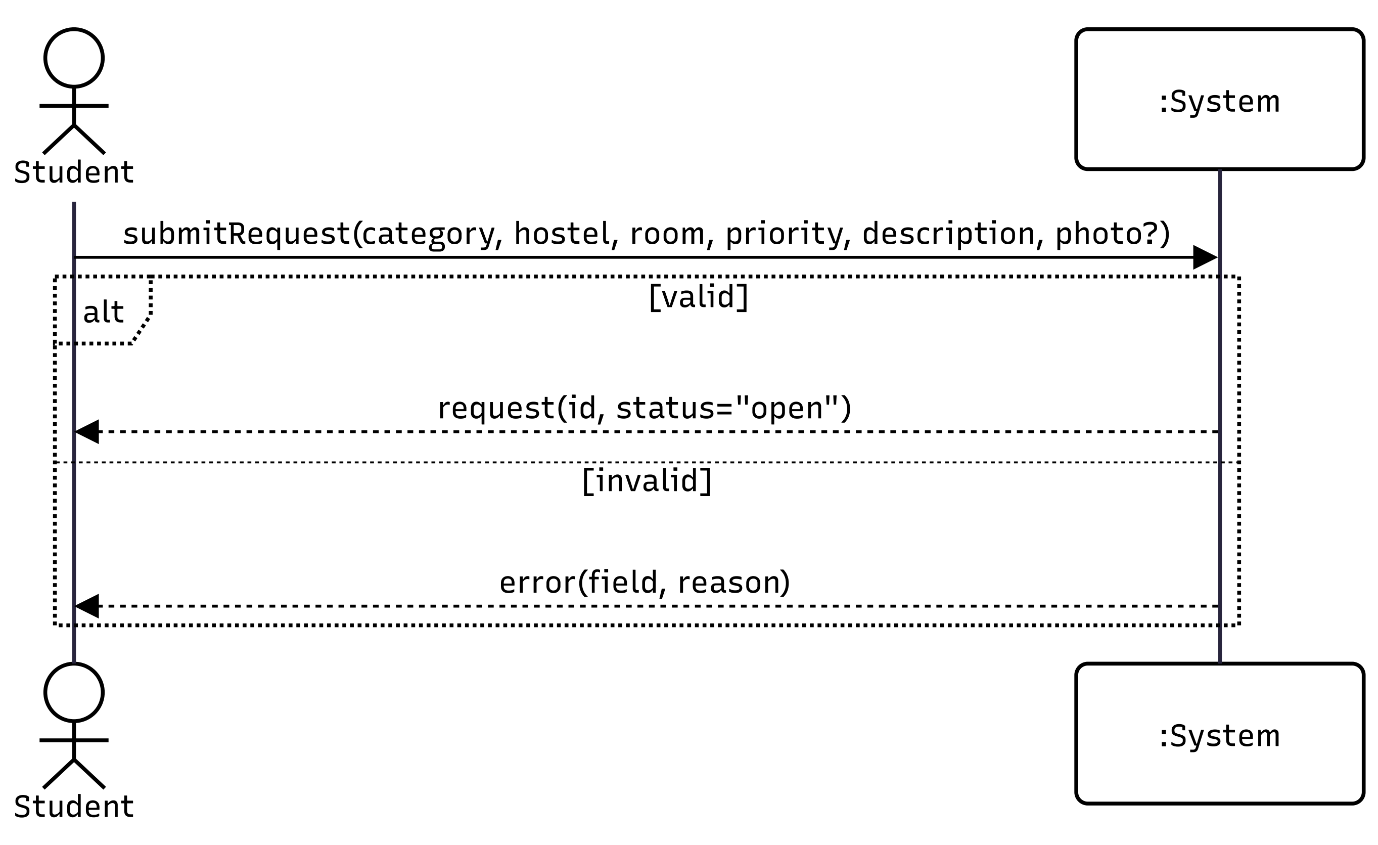
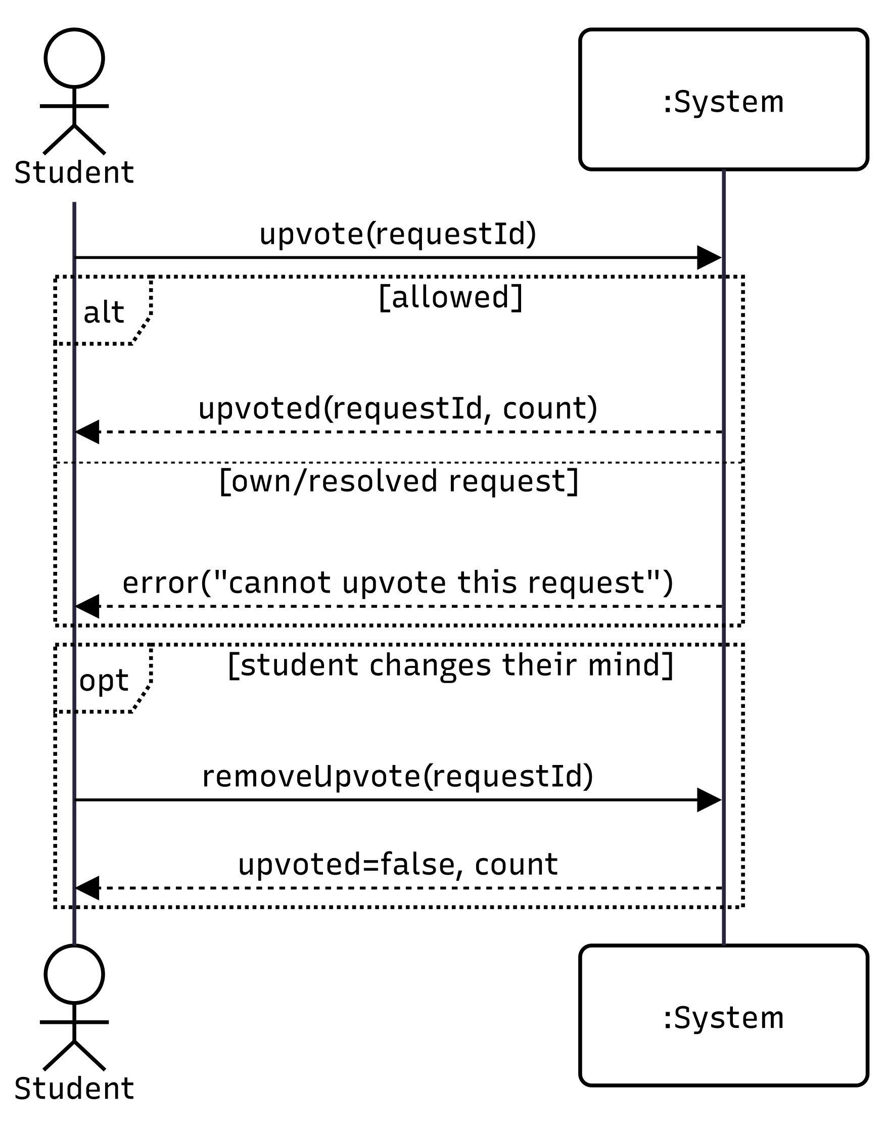
Use Case 2 - Log In


User
:System
login(idOrEmail, password)
alt
[credentials valid]
session(token), profile(role)
[invalid]
error("invalid credentials")
User
:System

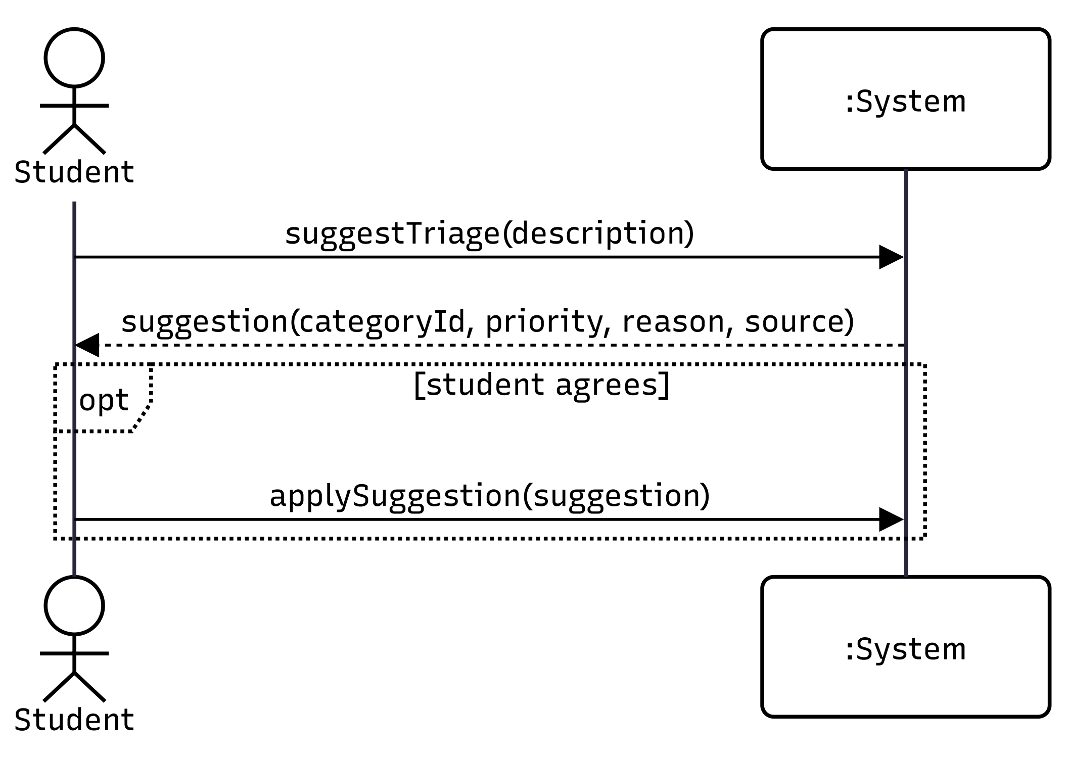
Use Case 3 - Submit a maintenance request


Student
:System
submitRequest(category, hostel, room, priority, description, photo?)
alt
[valid]
request(id, status="open")
[invalid]
error(field, reason)
Student
:System

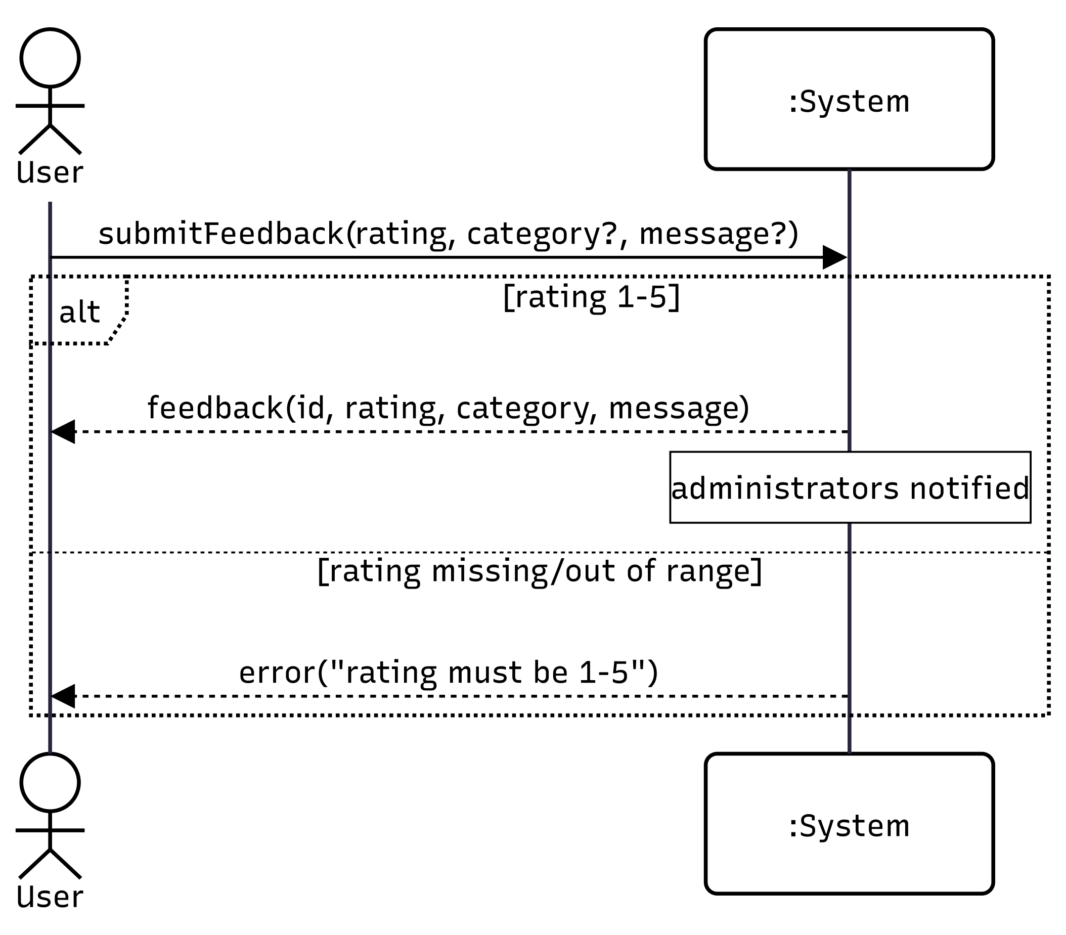
Use Case 6 - Back an existing report ("Me too / Upvote")



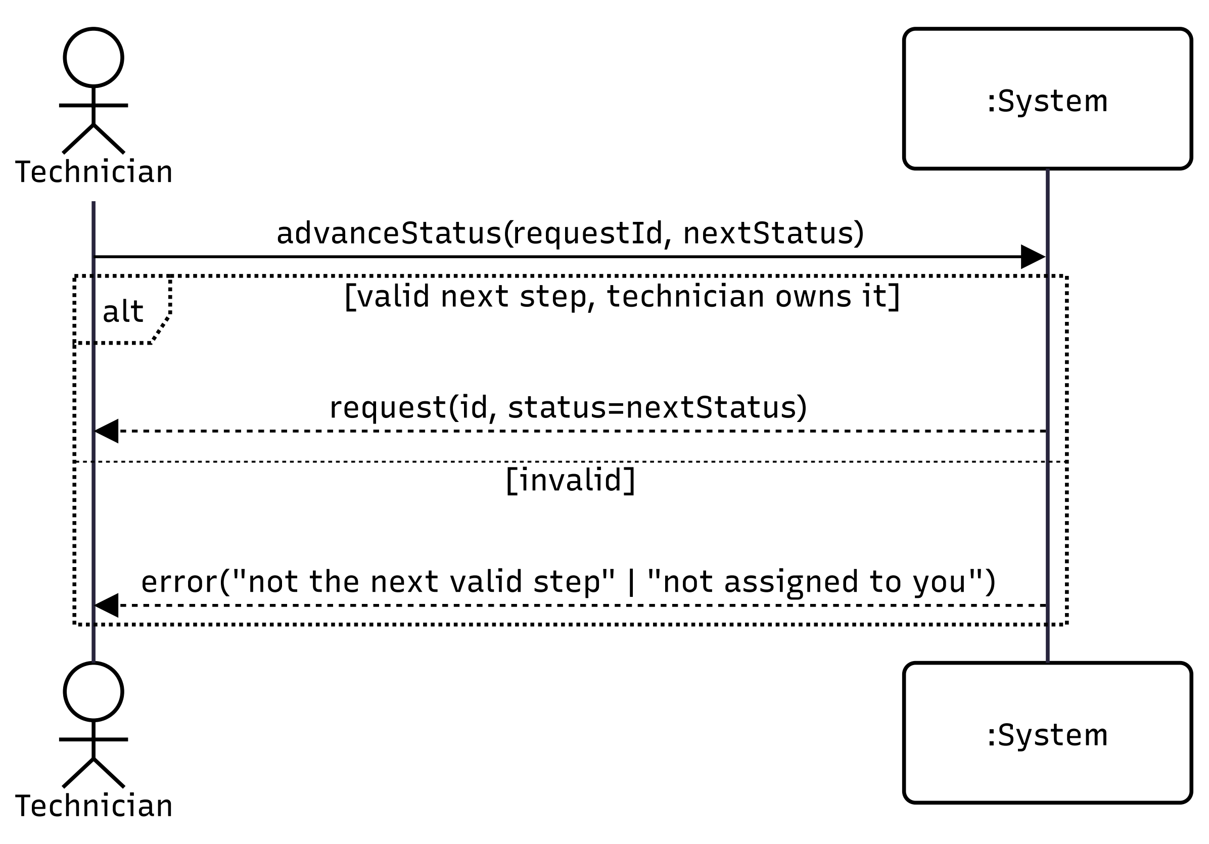
Use Case 9 - Get a triage suggestion


Student
:System
suggestTriage(description)
suggestion(categoryId, priority, reason, source)
opt
[student agrees]
applySuggestion(suggestion)
Student
:System

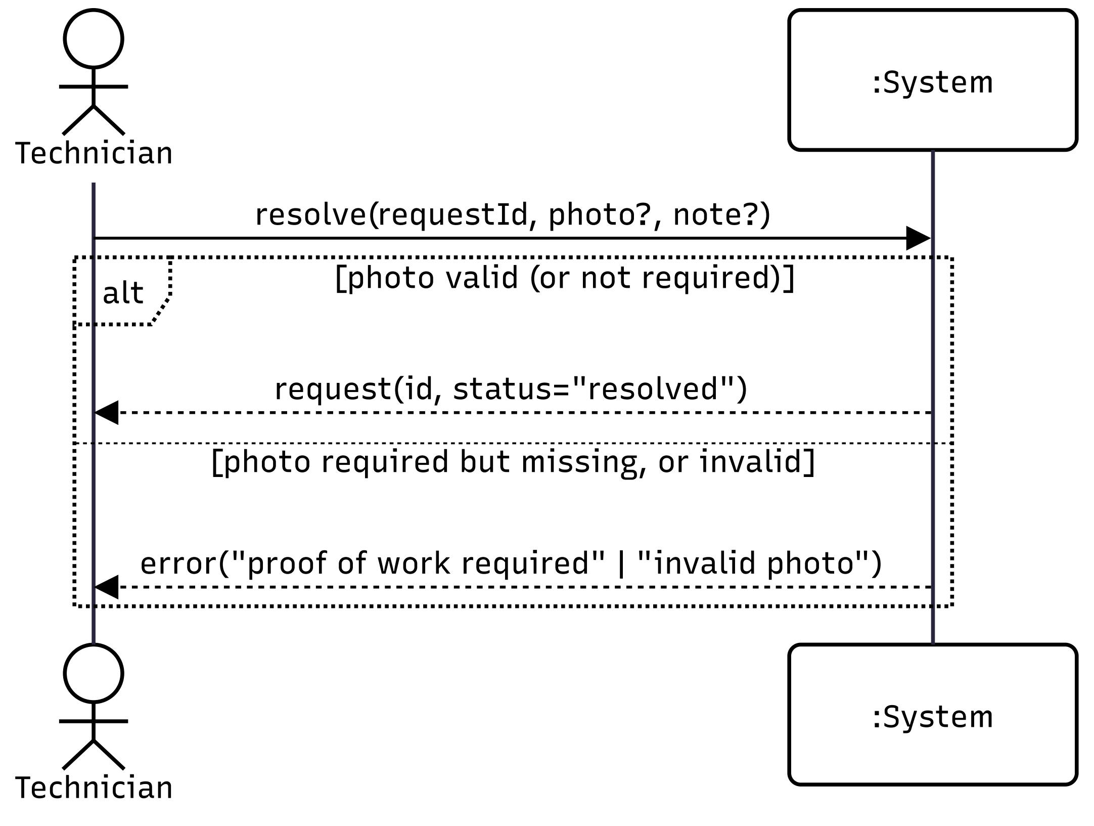
Use Case 13 - Submit app feedback


User
:System
submitFeedback(rating, category?, message?)
alt
[rating 1-5]
feedback(id, rating, category, message)
administrators notified
[rating missing/out of range]
error("rating must be 1-5")
User
:System

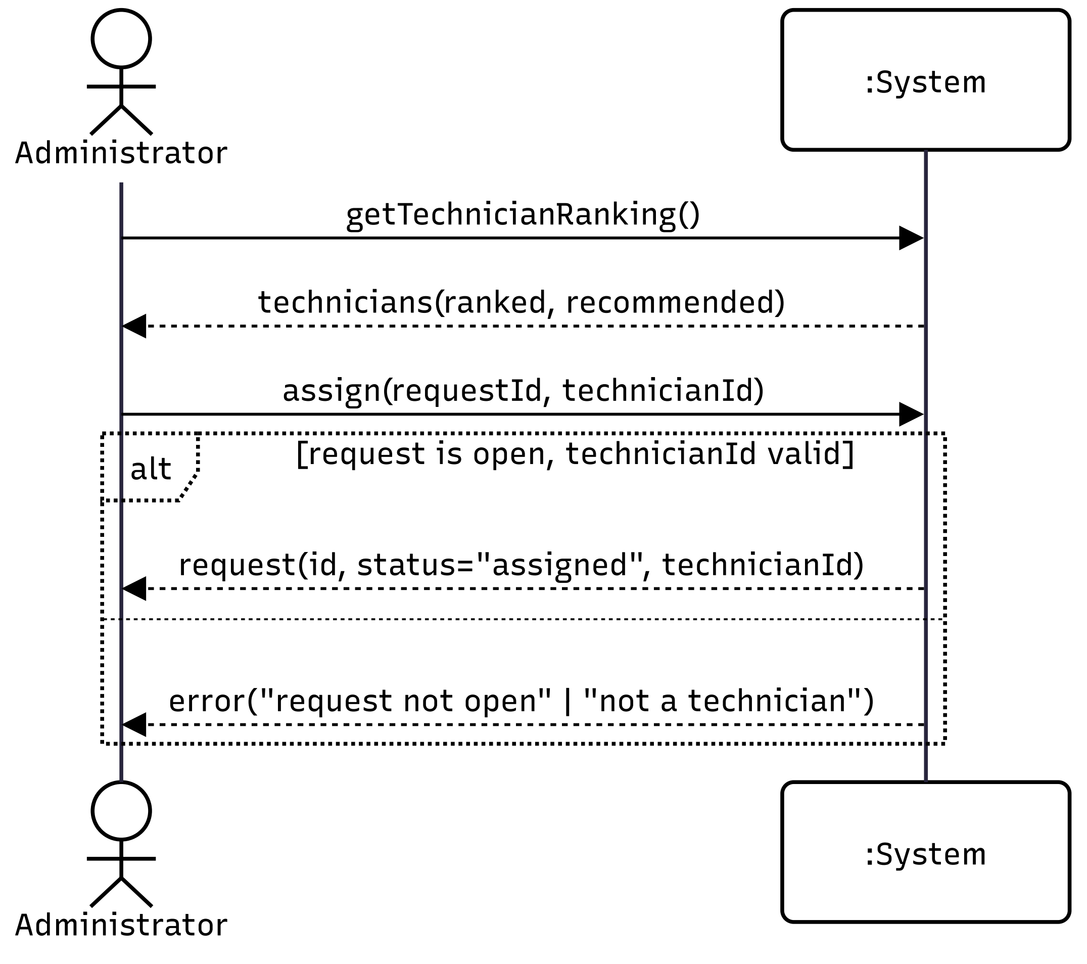
Use Case 16 - Update the status of an assigned request


Technician
:System
advanceStatus(requestId, nextStatus)
alt
[valid next step, technician owns it]
request(id, status=nextStatus)
[invalid]
error("not the next valid step" | "not assigned to you")
Technician
:System

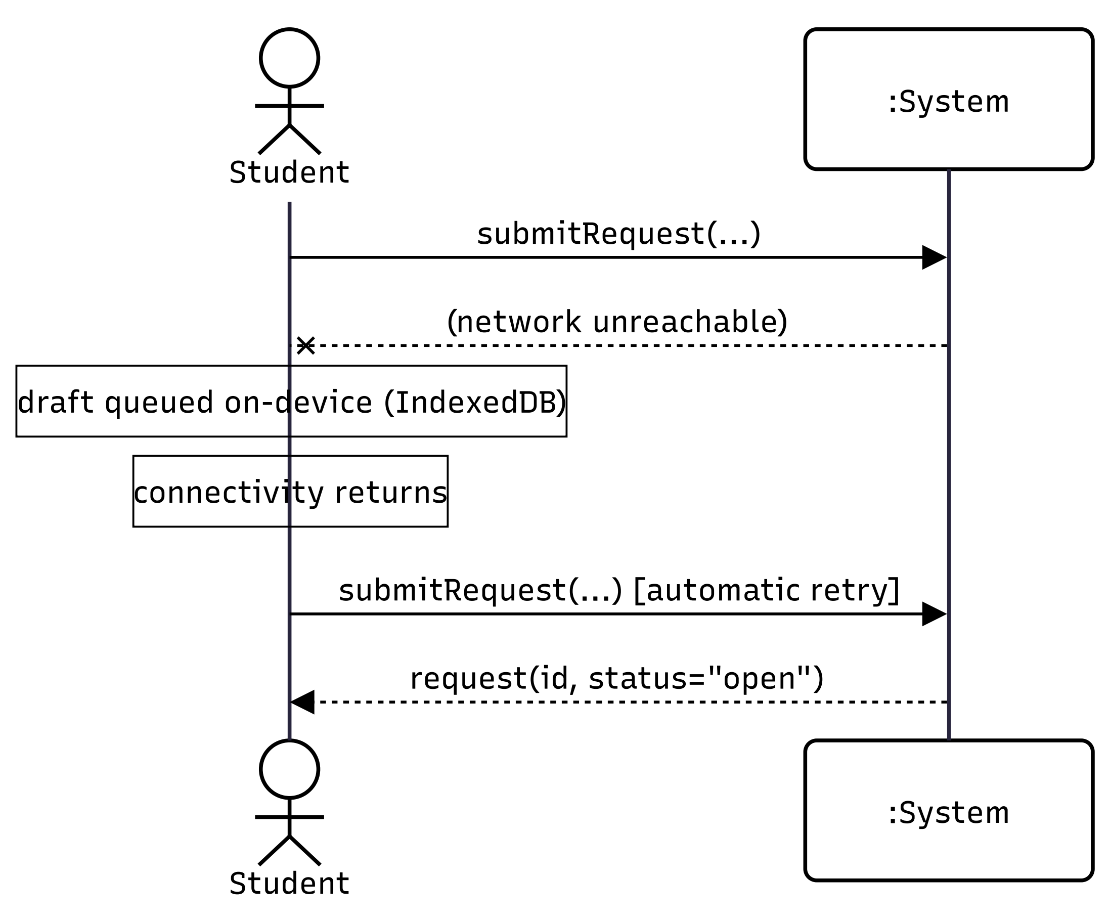
Use Case 17 - Attach proof of finished work


Technician
:System
resolve(requestId, photo?, note?)
alt
[photo valid (or not required)]
request(id, status="resolved")
[photo required but missing, or invalid]
error("proof of work required" | "invalid photo")
Technician
:System

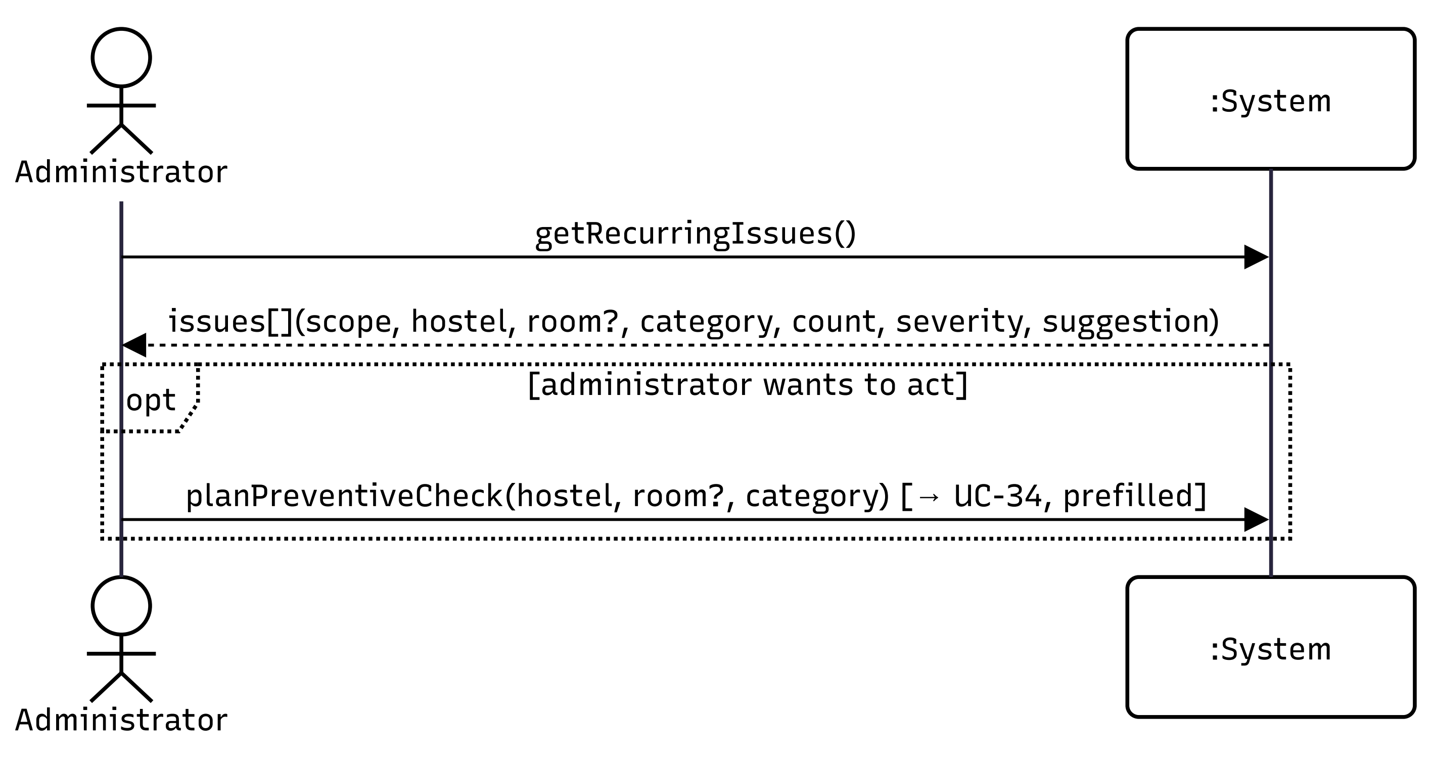
Use Case 23 - Assign a technician to a request


Administrator
:System
getTechnicianRanking()
technicians(ranked, recommended)
assign(requestId, technicianId)
alt
[request is open, technicianId valid]
request(id, status="assigned", technicianId)
error("request not open" | "not a technician")
Administrator
:System

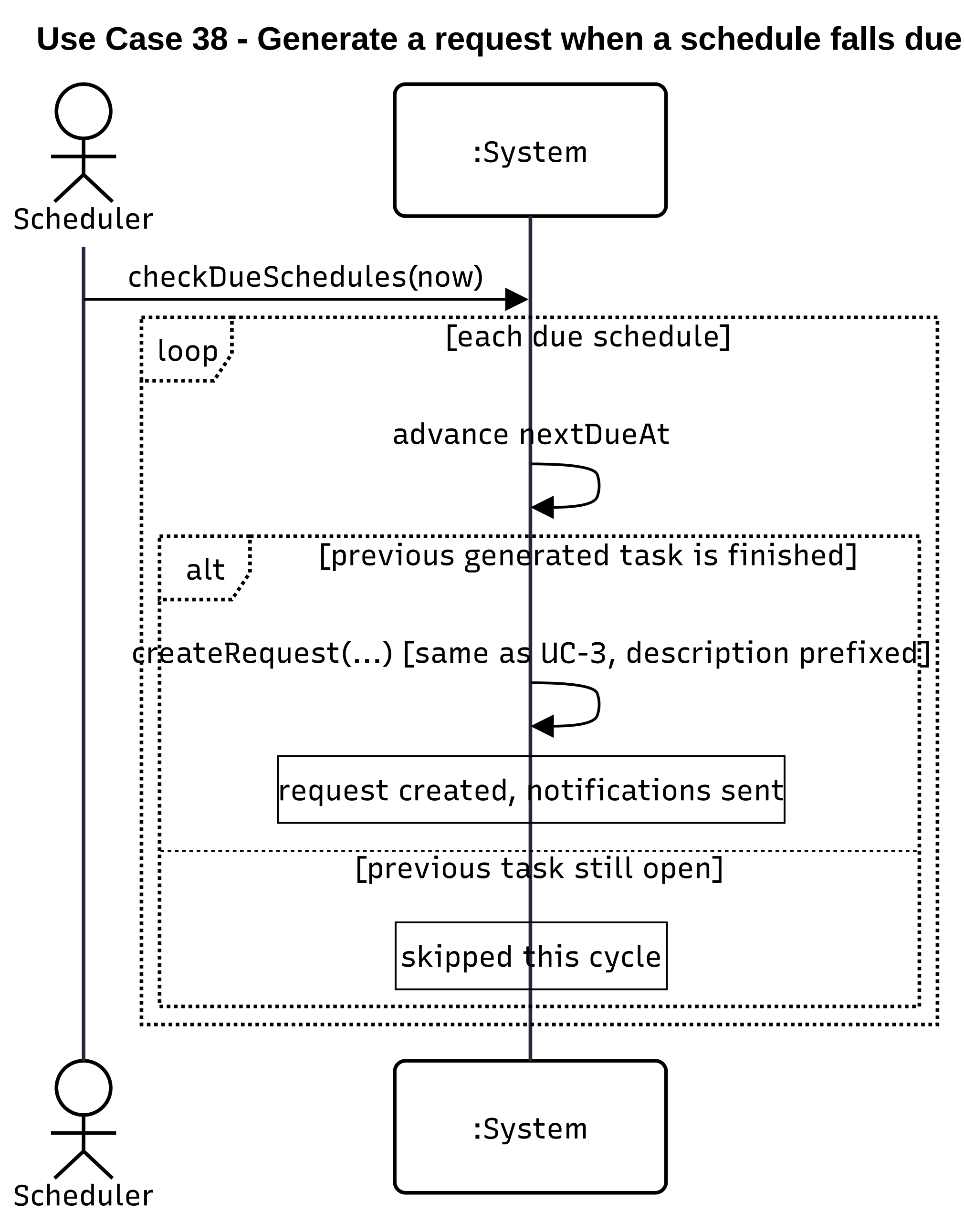
Use Case 10 - Draft a request offline


Student
:System
submitRequest(...)
(network unreachable)
draft queued on-device (IndexedDB)
connectivity returns
submitRequest(...) [automatic retry]
request(id, status="open")
Student
:System

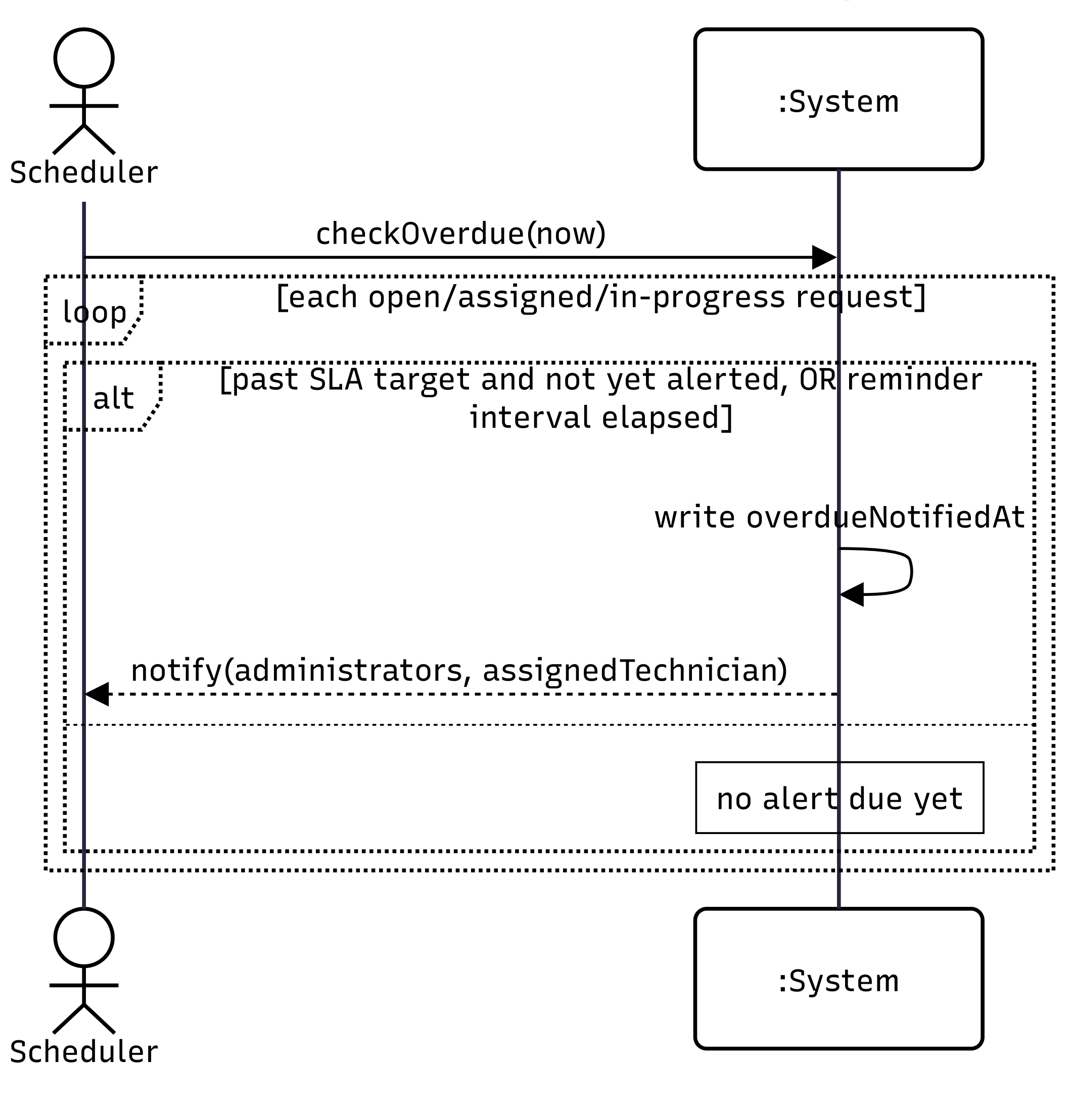
Use Case 31 - View recurring-issue detection


Administrator
:System
getRecurringIssues()
issues[](scope, hostel, room?, category, count, severity, suggestion)
opt
[administrator wants to act]
planPreventiveCheck(hostel, room?, category) [→ UC-34, prefilled]
Administrator
:System

Use Case 38 - Generate a request when a schedule falls due


Scheduler
:System
checkDueSchedules(now)
loop
[each due schedule]
advance nextDueAt
alt
[previous generated task is finished]
createRequest(...) [same as UC-3, description prefixed]
request created, notifications sent
[previous task still open]
skipped this cycle
Scheduler
:System

Use Case 39 - Detect and alert on an overdue request


Scheduler
:System
checkOverdue(now)
loop
[each open/assigned/in-progress request]
alt
[past SLA target and not yet alerted, OR reminder interval elapsed]
write overdueNotifiedAt
notify(administrators, assignedTechnician)
no alert due yet
Scheduler
:System